

هوش مصنوعی و کارگاه

دانشکده ریاضی و علوم کامپیوتر دانشگاه صنعتی امیرکبیر



مدرس کلاس: مهدی قطعی

استاد گروه علوم کامپیوتر

www.aut.ac.ir

مدرس کارگاه: بهنام یوسفی مهر

دانشجوی دکتری علوم کامپیوتر

Artificial Intelligence & Workshop

انتخاب ویژگی و استخراج ویژگی

انتخاب ویژگی و استخراج ویژگی

- در یادگیری ماشین، ویژگی‌ها (Features) متغیرهای ورودی هستند که مدل برای یادگیری الگوها و پیش‌بینی‌ها از آن‌ها استفاده می‌کند. کیفیت و مرتبط بودن این ویژگی‌ها تأثیر مستقیمی بر عملکرد مدل دارد.
- دو فرآیند اصلی برای بهبود کیفیت ویژگی‌ها وجود دارد: انتخاب ویژگی (Feature Selection) و استخراج ویژگی (Feature Extraction).
- انتخاب ویژگی به معنی انتخاب زیرمجموعه‌ای از ویژگی‌های مرتبط از داده‌های اصلی است، در حالی که استخراج ویژگی ایجاد ویژگی‌های جدید از طریق تبدیل داده‌های اصلی است.
- این دو روش به کاهش ابعاد داده، بهبود دقت مدل، کاهش زمان آموزش و جلوگیری از بیش‌برازش کمک می‌کنند .

تفاوت‌های انتخاب ویژگی و استخراج ویژگی

جنبه	انتخاب ویژگی	استخراج ویژگی
تعریف	انتخاب زیرمجموعه‌ای از ویژگی‌های مرتبط از مجموعه اصلی	تبدیل ویژگی‌های اصلی به یک فضای ویژگی جدید با ابعاد کمتر
هدف	کاهش ابعاد با حذف ویژگی‌های نامرتبط و افزونه	کاهش ابعاد با ایجاد ویژگی‌های جدید اطلاعاتی و فشرده
فرآیند	فیلتر کردن، الگوریتم‌های پوششی و تعبیه‌شده	تبدیل‌های خطی و غیرخطی، تکنیک‌های آماری، پردازش سیگنال و تصویر
خروجی	زیرمجموعه‌ای از ویژگی‌های اصلی	مجموعه‌ای جدید از ویژگی‌های تبدیل‌شده
هزینه محاسباتی	معمولاً پایین	ممکن است بالا باشد، به ویژه برای تبدیل‌های پیچیده
تفسیرپذیری	بالا (ویژگی‌های اصلی حفظ می‌شوند)	ممکن است پایین باشد (ویژگی‌های جدید ممکن است تفسیرپذیر نباشند)
حفظ ساختار داده	بله	خیر (ویژگی‌های جدید ایجاد می‌شوند)

روشهای انتخاب ویژگی

■ روشهای فیلتری (Filter)

- The filter methods pick up the intrinsic properties of the features (i.e., the “relevance” of the features) measured via univariate statistics instead of cross-validation performance.

■ الگوریتمهای پوششی (Wrapper)

- Wrapper methods measure the “usefulness” of features based on the classifier performance. So, wrapper methods essentially solve the “real” problem (optimizing classifier performance), but they are also computationally more expensive compared to filter methods due to the repeated learning steps and cross-validation.

■ الگوریتمهای تعبیه شده (Embedded) یا دورگه ای (Hybrid)

- The embedded methods are quite similar to wrapper methods since they are also used to optimize the objective function or performance of a learning algorithm or model. The difference to wrapper methods is that an intrinsic model-building metric is used during learning.

روشهای انتخاب ویژگی: اقدامات لازم

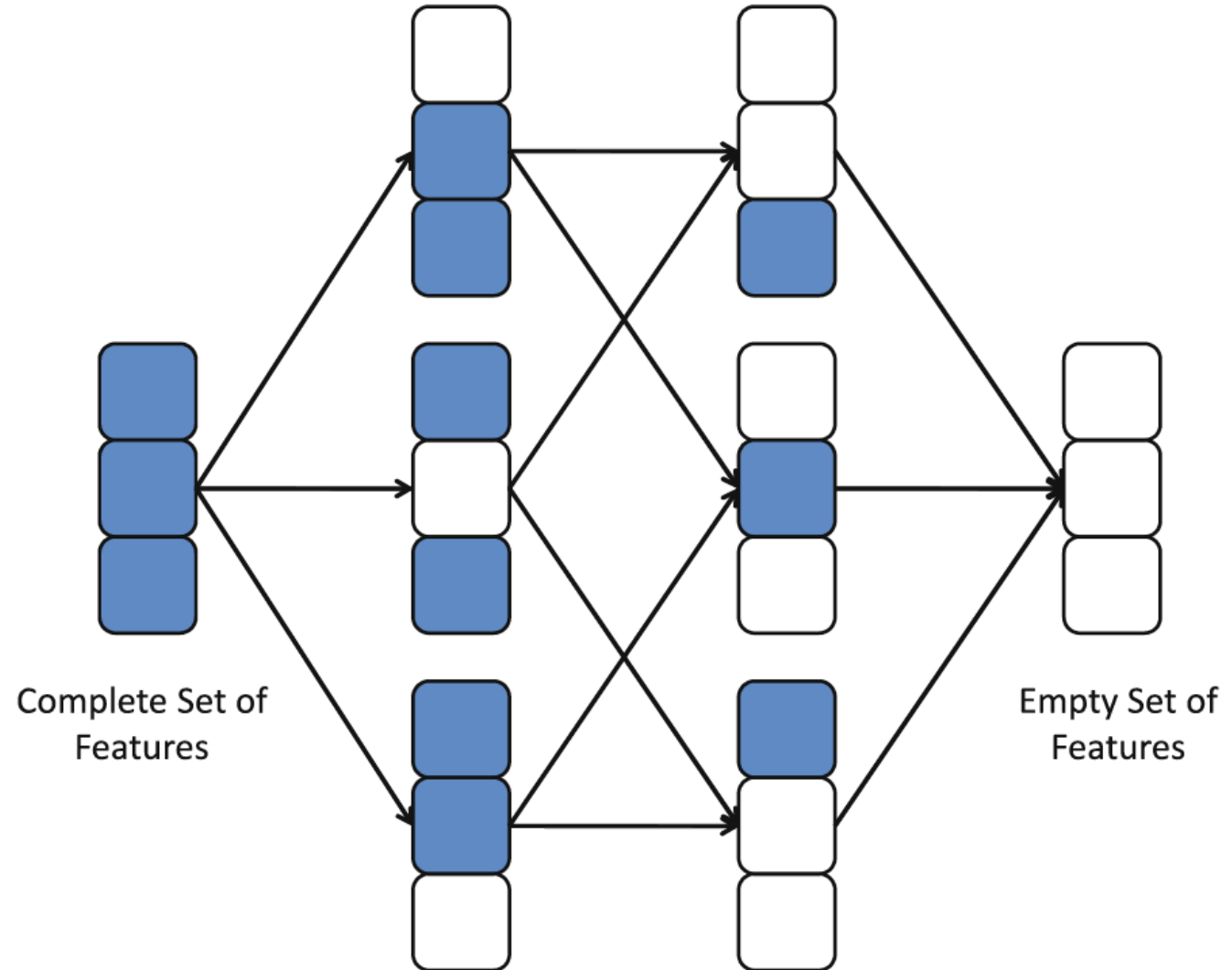
1. Searching for the best subset of features.
2. Criteria for evaluating different subsets.
3. Principles for selecting, adding, removing or changing new features during the search.

Search of a Subset of Features

- FS can be considered as a search problem, where each state of the search space corresponds to a concrete subset of features selected.
- The selection can be represented as a binary array, with each element corresponding to the value 1, if the feature is currently selected by the algorithm and 0, if it does not occur.
- There should be a total of 2^M subsets where M is the number of features of a data set.

Search of a Subset of Features

Search
Space:



Search of a Subset of Features

- **Search Directions:**

- **Sequential Forward Generation (SFG):** It starts with an empty set of features S . As the search starts, features are added into S according to some criterion that distinguish the best feature from the others. S grows until it reaches a full set of original features. The stopping criteria can be a threshold for the number of relevant features m or simply the generation of all possible subsets in brute force mode.
- **Sequential Backward Generation (SBG):** It starts with a full set of features and, iteratively, they are removed one at a time. Here, the criterion must point out the worst or least important feature. By the end, the subset is only composed of a unique feature, which is considered to be the most informative of the whole set. As in the previous case, different stopping criteria can be used.

Search of a Subset of Features

- Search Directions:

Algorithm 1 Sequential forward feature set generation - SFG.

function SFG(F - full set, U - measure)

initialize: $S = \{\}$

 ▷ S stores the selected features

repeat

$f = \text{FINDNEXT}(F)$

$S = S \cup \{f\}$

$F = F - \{f\}$

until S satisfies U or $F = \{\}$

return S

end function

Search of a Subset of Features

- Search Directions:

Algorithm 2 Sequential backward feature set generation - SBG.

function SBG(F - full set, U - measure)

initialize: $S = \{\}$

 ▷ S holds the removed features

repeat

$f = \text{GETNEXT}(F)$

$F = F - \{f\}$

$S = S \cup \{f\}$

until S does not satisfy U or $F = \{\}$

return $F \cup \{f\}$

end function

Search of a Subset of Features

- **Search Directions:**

- **Bidirectional Generation (BG):** Begins the search in both directions, performing SFG and SBG concurrently. They stop in two cases: (1) when one search finds the best subset comprised of m features before it reaches the exact middle, or (2) both searches achieve the middle of the search space. It takes advantage of both SFG and SBG.
- **Random Generation (RG):** It starts the search in a random direction. The choice of adding or removing a features is a random decision. RG tries to avoid the stagnation into a local optima by not following a fixed way for subset generation. Unlike SFG or SBG, the size of the subset of features cannot be stipulated.

Search of a Subset of Features

■ Search Directions:

Algorithm 3 Bidirectional feature set generation - BG.

function BG(F_f , F_b - full set, U - measure)

initialize: $S_f = \{\}$

▷ S_f holds the selected features

initialize: $S_b = \{\}$

▷ S_b holds the removed features

repeat

$f_f = \text{FINDNEXT}(F_f)$

$f_b = \text{GETNEXT}(F_b)$

$S_f = S_f \cup \{f_f\}$

$F_b = F_b - \{f_b\}$

$F_f = F_f - \{f_f\}$

$S_b = S_b \cup \{f_b\}$

until (a) S_f satisfies U or $F_f = \{\}$ or (b) S_b does not satisfy U or $F_b = \{\}$

return S_f if (a) or $F_b \cup \{f_b\}$ if (b)

end function

Search of a Subset of Features

■ Search Directions:

Algorithm 4 Random feature set generation - RG.

function RG(F - full set, U - measure)

initialize: $S = S_{best} = \{\}$

▷ S - subset set

initialize: $C_{best} = \#(F)$

▷ $\#$ - cardinality of a set

repeat

$S = \text{RANDGEN}(F)$

$C = \#(S)$

if $C \leq C_{best}$ and S satisfies U **then**

$S_{best} = S$

$C_{best} = C$

end if

until some stopping criterion is satisfied

return S_{best}

▷ Best set found so far

end function

Search of a Subset of Features

■ Search Strategies:

- **Exhaustive Search:** It corresponds to explore all possible subsets to find the optimal ones. As we said before, the space complexity is $O(2^M)$. If we establish a threshold m of minimum features to be selected and the direction of search, the search space is, independent of the forward or backward generation. Only exhaustive search can guarantee the optimality. Nevertheless, they are also impractical in real data sets with a high M .
- **Heuristic Search:** It employs heuristics to carry out the search. Thus, it prevents brute force search, but it will surely find a non-optimal subset of features. It draws a path connecting the beginning and the end of the previous Figure, such in a way of a depth-first search. The maximum length of this path is M and the number of subsets generated is $O(M)$. The choice of the heuristic is crucial to find a closer optimal subset of features in a faster operation.

Selection Criteria

- Information Measures.
 - Information serves to measure the uncertainty of the receiver when she/he receives a message.
 - Shannon's Entropy:

$$-\sum_i P(c_i) \log_2 P(c_i).$$

- Information gain:

$$IG(A) = I(D) - \sum_{j=1}^p \frac{|D_j|}{|D|} I(D_j^A)$$

آنتروپی شانون چه چیزی را اندازه گیری می کند؟

- آنتروپی شانون معیاری برای عدم قطعیت یا اطلاعات میانگین در یک سیستم است. به زبان ساده:
- آنتروپی بالا = عدم قطعیت بالا = بینظمی بالا = غیرقابل پیشبینی بودن
- آنتروپی پایین = عدم قطعیت پایین = نظم بالا = قابل پیشبینی بودن
- مثال: دو آزمایش را در نظر بگیرید:
- آزمایش A (کمتر تصادفی): یک سکهی تقلبی که ۹۰٪ مواقع "شیر" و ۱۰٪ مواقع "خط" می آید: محاسبه آنتروپی:
$$H(A) = - [0.9 * \log_2(0.9) + 0.1 * \log_2(0.1)] \approx 0.47$$
- آزمایش B (بسیار تصادفی): یک سکه کاملاً منصفانه که ۵۰٪ مواقع "شیر" و ۵۰٪ مواقع "خط" می آید: محاسبه آنتروپی:
$$H(B) = - [0.5 * \log_2(0.5) + 0.5 * \log_2(0.5)] = 1$$
- نتیجه: آنتروپی آزمایش B به وضوح از آنتروپی آزمایش A بیشتر است.
- چرا اینطور است؟
- در آزمایش A سکه تقلبی، شما تقریباً مطمئن هستید که نتیجه "شیر" خواهد بود. بنابراین عدم قطعیت و غافلگیری کمی وجود دارد. وقتی پیام "شیر" را دریافت می کنید، اطلاعات جدید زیادی کسب نکرده اید زیرا از قبل آن را پیشبینی میکردید.

Perspectives: Selection Criteria

- Distance Measures.
 - Measures of separability, discrimination or divergence measures . The most typical is derived from distance between the class conditional density functions.

	Mathematical form
Euclidean distance	$D_e = \{\sum_{i=1}^m (x_i - y_i)^2\}^{\frac{1}{2}}$
City-block distance	$D_{cb} = \sum_{i=1}^m x_i - y_i $
Cebyshev distance	$D_{ch} = \max_i x_i - y_i $
Minkowski distance of order m	$D_M = \{\sum_{i=1}^m (x_i - y_i)^m\}^{\frac{1}{m}}$
Quadratic distance Q , positive definite	$D_q = \sum_{i=1}^m \sum_{j=1}^m (x_i - y_i) Q_{ij} (x_j - y_j)$
Canberra distance	$D_{ca} = \sum_{i=1}^m \frac{ x_i - y_i }{x_i + y_i}$
Angular separation	$D_{as} = \frac{\sum_{i=1}^m x_i \cdot y_i}{[\sum_{i=1}^m x_i^2 \sum_{i=1}^m y_i^2]^{\frac{1}{2}}}$

Perspectives: Selection Criteria

- Dependence Measures.
 - known as measures of association or correlation.
 - Its main goal is to quantify how strongly two variables are correlated or present some association with each other, in such way that knowing the value of one of them, we can derive the value for the other.
 - *Pearson correlation* coefficient:

$$\rho(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\left[\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2 \right]^{\frac{1}{2}}}$$

Perspectives: Selection Criteria

- Consistency Measures.
 - They attempt to find a minimum number of features that separate classes as the full set of features can.
 - They aim to achieve $P(C | \text{FullSet}) = P(C | \text{SubSet})$.
 - An inconsistency is defined as the case of two examples with the same inputs (same feature values) but with different output feature values (classes in classification).

Perspectives: Selection Criteria

■ Accuracy Measures.

- This form of evaluation relies on the classifier or learner. Among various possible subsets of features, the subset which yields the best predictive accuracy is chosen

	Mathematical form
Accuracy	$\frac{tp+fp}{tp+tn+fp+fn}$
Error rate	1 – Accuracy
Chi-squared	$\frac{n(fp \times fn - tp \times tn)^2}{(tp+fp)(tp+fn)(fp+tn)(tn+fn)}$
Information gain	$e(tp + fn, fp + tn) - \frac{(tp+fp)e(tp,fp)+(tn+fn)e(fn,tn)}{tp+fp+tn+fn}$ where $e(x, y) = -\frac{x}{x+y} \log_2 \frac{x}{x+y} - \frac{y}{x+y} \log_2 \frac{y}{x+y}$
Odds ratio	$\frac{tpr}{1-tpr} \bigg/ \frac{fpr}{1-fpr} = \frac{tp \times tn}{fp \times fn}$
Probability ratio	$\frac{tpr}{fpr}$

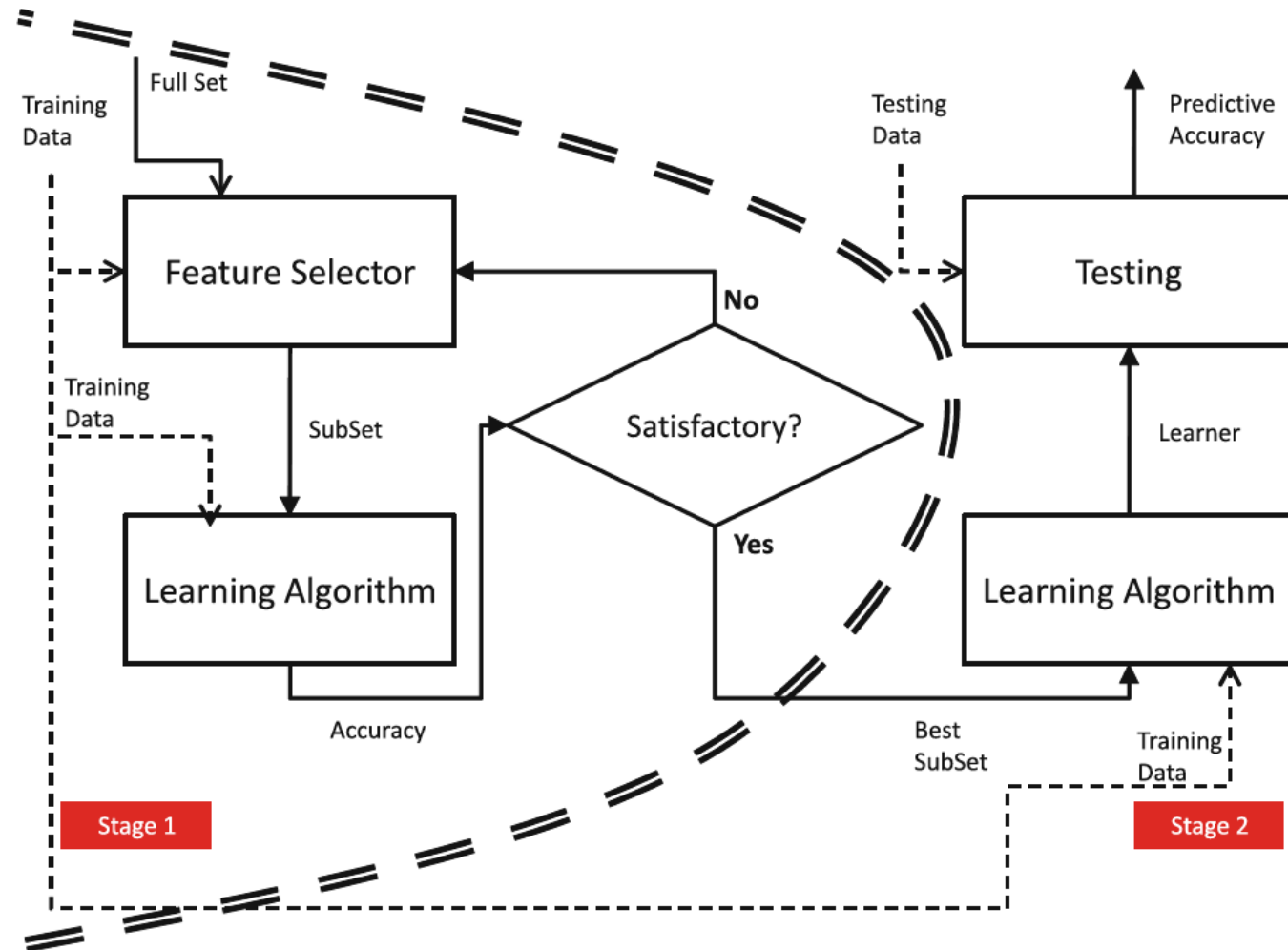
Perspectives

- Filters:

- measuring uncertainty, distances, dependence or consistency is usually cheaper than measuring the accuracy of a learning process. Thus, filter methods are usually faster.
- it does not rely on a particular learning bias, in such a way that the selected features can be used to learn different models from different DM techniques.
- it can handle larger sized data, due to the simplicity and low time complexity of the evaluation measures.

Perspectives

- Wrappers:



Perspectives

- Wrappers:
 - can achieve the purpose of improving the particular learner's predictive performance.
 - usage of internal statistical validation to control the overfitting, ensembles of learners and hybridizations with heuristic learning like Bayesian classifiers or Decision Tree induction.
 - filter models cannot allow a learning algorithm to fully exploit its bias, whereas wrapper methods do.

Perspectives

- Embedded FS:
 - similar to the wrapper approach in the sense that the features are specifically selected for a certain learning algorithm, but in this approach, the features are selected during the learning process.
 - they could take advantage of the available data by not requiring to split the training data into a training and validation set; they could achieve a faster solution by avoiding the re-training of a predictor for each feature subset explored.

Aspects:

Output of Feature Selection

- Feature Ranking Techniques:
 - we expect as the output a ranked list of features which are ordered according to evaluation measures.
 - they return the relevance of the features.
 - For performing actual FS, the simplest way is to choose the first m features for the task at hand, whenever we know the most appropriate m value.

Aspects: Output of Feature Selection

- Feature Ranking Techniques:

Algorithm 5 A univariate feature ranking algorithm.

```
function RANKING_ALGORITHM( $x$  - features,  $U$  - measure)
  initialize: list  $L = \{\}$                                      ▷  $L$  stores ordered features
  for each feature  $x_i, i \in \{1, \dots, M\}$  do
     $v_i = \text{COMPUTE}(x_i, U)$ 
    position  $x_i$  into  $L$  according to  $v_i$ 
  end for
  return  $L$  in decreasing order of feature relevance.
end function
```

Aspects:

Output of Feature Selection

- Minimum Subset Techniques:
 - The number of relevant features is a parameter that is often not known by the practitioner.
 - There must be a second category of techniques focused on obtaining the minimum possible subset without ordering the features.
 - whatever is relevant within the subset, is otherwise irrelevant.

Aspects: Output of Feature Selection

- Minimum Subset Techniques:

Algorithm 6 A minimum subset algorithm.

function MIN- SET ALGORITHM(x - features, U - measure)

initialize: $L = \{\}$, $\text{stop} = \text{false}$

▷ S holds the minimum set

repeat

$S_k = \text{SUBSETGENERATE}(x)$

▷ **stop** can be set here

if LEGITIMACY(S_k, U) is *true* and $\#(S_k) < \#(S)$ **then**

$S = S_k$

▷ S is replaced by S_k

end if

until $\text{stop} = \text{true}$

return S - the minimum subset of features

end function

Aspects: Evaluation

- **Goals:**

- **Inferability:** For predictive tasks, considered as an improvement of the prediction of unseen examples with respect to the direct usage of the raw training data.
- **Interpretability:** Given the incomprehension of raw data by humans, DM is also used for generating more understandable structure representation that can explain the behavior of the data.
- **Data Reduction:** It is better and simpler to handle data with lower dimensions in terms of efficiency and interpretability.

Aspects: Evaluation

- We can derive three assessment measures from these three goals:
 - Accuracy
 - Complexity
 - Number of Features Selected
 - Speed of the FS method
 - Generality of the features selected

Aspects: Drawbacks

- The resulted subsets of many models of FS are strongly dependent on the training set size.
- It is not true that a large dimensionality input can always be reduced to a small subset of features because the objective feature is actually related with many input features and the removal of any of them will seriously effect the learning performance.
- A backward removal strategy is very slow when working with large-scale data sets. This is because in the firsts stages of the algorithm, it has to make decisions funded on huge quantities of data.
- In some cases, the FS outcome will still be left with a relatively large number of relevant features which even inhibit the use of complex learning methods.

Aspects:

Using Decision Trees for FS

- Decision trees can be used to implement a trade-off between the performance of the selected features and the computation time which is required to find a subset.
- Decision tree inducers can be considered as anytime algorithms for FS, due to the fact that they gradually improve the performance and can be stopped at any time, providing sub-optimal feature subsets.

Ref:

- **Soft Computing and Intelligent Information Systems**
- A University of Granada research group
- <https://sci2s.ugr.es/>